

2nd Place

GLOBAL ASME IAM3D

40+ Int'l Universities

40%

PROVISIONING BOOST

Terraform & AWS IaC

150+

STUDENTS TAUGHT

Embedded & Firmware Labs

28%

CONVERGENCE SPEEDUP

PennyLane + AWS Braket

PROFESSIONAL EXPERIENCE

INDUSTRY & LEADERSHIP

Cloud Infrastructure Engineering Intern • University of Massachusetts Amherst IT

Sept 2025 – Present | Amherst, MA

- Architect and deploy automated cloud infrastructure and CI/CD pipelines across AWS using **Terraform**, reducing environment provisioning turnaround by **40%** and eliminating manual configuration drift.
- Develop telemetry, health-check, and monitoring services in **Python and Bash** to isolate platform faults and improve service uptime across university IT platforms.
- Write reusable Terraform modules and containerized microservices in **Docker** to standardize deployment setups across 6 engineering teams.

AWS (EC2, S3, Lambda, IAM)

Terraform

Docker

Python

Bash

CI/CD

Linux/Unix

Systems & Embedded Engineering Lead / Instructor • Stem Studio

May 2023 – Present | Chicago, IL

- Led emergency incident remediation for a server-side cloaking exploit and backdoor on production servers, restoring full system stability and uptime with zero data loss.
- Created and taught hands-on embedded computing curricula (microcontrollers, circuit design, sensor interfacing, and robotics) for **150+ students**, raising project pass rates by **35%**.
- Authored reproducible laboratory guides, hardware documentation, and sample firmware repositories.

Embedded C

ARM Cortex

ESP32 / Arduino

I2C / SPI / UART

Linux Hardening

Incident Response

Technical Mentor & Lead Instructor • CodeDay

Aug 2020 – Aug 2023 | Global Initiative

- Mentored **20+ student engineering teams** through complete project lifecycles, covering data structures, API integrations, algorithms, and Git collaboration.
- Conducted regular code reviews, debugging walkthroughs, and architecture reviews to help teams deploy functional software.

Python

JavaScript

REST APIs

Git / GitHub

Data Structures

EDUCATION & ACADEMIC LEADERSHIP

UMASS AMHERST

Bachelor of Science in Computer Engineering • University of Massachusetts Amherst

Graduation: May 2026 | Amherst, MA

College: Riccio College of Engineering | Subteam Lead: ASME IAM3D Mechatronics & U.S. DOE Collegiate Wind

Key Courses: Computer Architecture, Embedded Systems Design, SystemVerilog & Digital Logic, Data Structures & Algorithms, Signals & Systems, Power Electronics.

Master of Science in Business Analytics (MSBA) • University of Massachusetts Amherst

Graduation: May 2027 | Amherst, MA

School: Isenberg School of Management | Focus on quantitative modeling, high-volume processing, and enterprise data architectures.

TECHNICAL COMPETENCIES

CORE SKILLS MATRIX

Programming & Systems

Languages: Python, JavaScript (ES6+), SystemVerilog, SQL, Java, Bash/Shell, C/C++

Quantum: Qiskit, PennyLane, Circuit Optimization, State-Vector Simulation

Embedded & Hardware

Firmware & MCUs: Embedded C, ARM Cortex, ESP32, Arduino, I2C / SPI / UART

EDA & Prototyping: LTSpice Simulation, SolidWorks CAD, CNC Machining, 3D Printing

Cloud & Infrastructure

Cloud Platforms: AWS (EC2, S3, Lambda, IAM, Braket), Google Cloud (GCP)

DevOps & Tools: Terraform (IaC), Docker, CI/CD Pipelines, Linux/Unix Internals

AI, ML & Analytics

Deep Learning: PyTorch, Variational Quantum Circuits (QNN), CNNs, NumPy

Analytics: High-Volume Processing, Predictive Modeling, REST APIs

Quantum-Classical Hybrid Neural Network

[GitHub Code ↗](#)

Variational Quantum Circuits on AWS Braket with PennyLane • Lead Architect & Developer

Designed and implemented a hybrid neural network integrating classical convolutional layers with parameterized variational quantum circuits executed on AWS Braket state-vector simulators, achieving 28% faster epoch convergence and 3.4x parameter efficiency.

28% Faster Epoch Convergence vs Classical CNN	3.4x Parameter Efficiency in Hilbert Space	AWS Braket Distributed Quantum Simulation
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Key Highlights: Custom PennyLane QNode integrated with PyTorch nn.Module forward loops; angle embedding with multi-qubit entanglement (CNOT + LCU); parameter-shift gradient computation; gate depth optimization.

[PyTorch](#) [PennyLane](#) [AWS Braket](#) [Python](#) [Qiskit](#) [NumPy](#)

🏆 2ND PLACE GLOBALLY • ASME IAM3D COMPETITION (40+ UNIVERSITIES)

Autonomous Payload Drone & Off-Terrain Rover

[GitHub Code ↗](#)

ASME UMass Amherst Mechatronics Team • Electrical & Prototyping Subteam Lead

Led electrical and prototyping subteam to engineer an articulated payload drone and off-terrain rover. Modeled custom CNC aluminum 6061 frame in SolidWorks, programmed ARM Cortex motor telemetry, and secured 2nd Place globally among 40+ international teams.

2nd Place Global ASME IAM3D Finish	4.5 kg Payload Capacity with FEA Tuning	50 Hz Real-time Telemetry over SPI/I2C
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[SolidWorks](#) [CNC Machining](#) [ARM Cortex](#) [Embedded C](#) [3D Printing](#) [I2C/SPI/UART](#) [Motor Drivers](#)

⚡ 100% DOE ELECTRICAL SAFETY COMPLIANCE

Offshore Wind Turbine 24V Power & Telemetry System

[GitHub Code ↗](#)

U.S. Department of Energy Collegiate Wind Competition • Electrical Subteam Lead (6 Engineers)

Led a 6-engineer subteam in modeling, simulating, and validating a 24V synchronous DC-DC power regulation circuit and sensor telemetry platform for dynamic wind-load conditions, achieving 100% safety and specification compliance.

24V ±0.5% Voltage Regulation Stability	100% DOE Benchmark Compliance	6 Engineers Led Schematic & Assembly
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[LTSpice](#) [Power Electronics](#) [24V DC-DC](#) [PCB Simulation](#) [Embedded C](#) [Telemetry](#)

Multi-Tenant Cloud Infrastructure & Terraform Automation

[GitHub Code ↗](#)

UMass Amherst IT • Cloud Infrastructure Engineering Intern

Built modular Terraform blueprints and automated CI/CD pipelines on AWS, accelerating developer onboarding and cloud environment provisioning by 40% with zero configuration drift across 6 engineering teams.

[AWS \(EC2, S3, Lambda, IAM\)](#) [Terraform](#) [Docker](#) [Python](#) [Bash](#) [CI/CD](#)

Embedded IoT Learning Platform

[GitHub ↗](#)

Stem Studio • Systems Lead

Created hands-on embedded firmware lab curriculum (ESP32, ARM Cortex, I2C/SPI) for 150+ students, raising project pass rates by 35%.

[Embedded C](#) [ARM](#) [ESP32](#) [IoT](#)

Malware Remediation & Hardening

[GitHub ↗](#)

Production Incident Response

Neutralized obfuscated PHP cloaking malware, restored WSOD failures, and implemented fail2ban and WAF security locks.

[Linux Hardening](#) [PHP](#) [Nginx](#) [Bash](#)